# Asymmetric Batch-One Training Learns Symmetric Functions in Low-Rank Weight Space

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## Abstract

We study an empirical phenomenon in low-rank random-feature networks: even when stochastic training sees one non-symmetrized sample at a time, the learned model can recover a globally symmetric function and, in successful regimes, symmetric internal partial functions. The effect is not explained by output invariance alone. Low-loss counterexamples exist where the output is nearly symmetric but the learned partial functions remain asymmetric. We therefore analyze symmetry simultaneously in function space, representation space, and weight space. Across one-dimensional ICLR sumcos reruns and new multidimensional batch-size-one experiments, low-rank bottleneck networks exhibit a distinctive regime where symmetry is visible in late partial functions and in mirror-paired first-layer atoms.

## 1. Question

Weight-space symmetries are usually discussed as exact parameter transformations that preserve a network function, especially in model merging and mode connectivity (Garipov et al., 2018; Frankle et al., 2020; Entezari et al., 2022; Ainsworth et al., 2023). Here we ask a different question: can optimization discover a *data-dependent* symmetry in weight space when the training procedure itself is maximally asymmetric? Our motivating setting uses a target invariant under $x \mapsto -x$ or related finite transformations, but trains with stochastic batch size one and without paired samples. The surprising observation is that MMNN/RF-LR models often learn not only an invariant output but also invariant low-rank partial functions.

For a learned scalar function $f$, we measure

$$D_{\text{out}}(T) = \frac{\mathbb{E}_x\left[(f(x) - f(Tx))^2\right]}{\mathbb{E}_x\left[f(x)^2\right] + \varepsilon},$$

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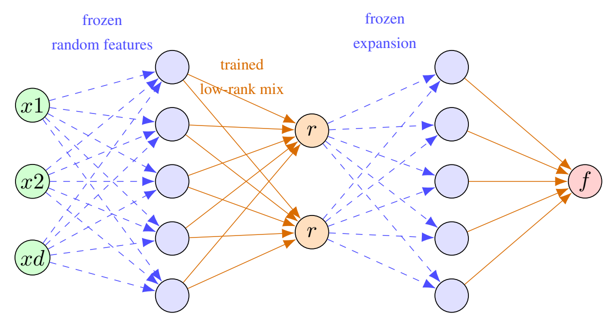


*Figure 1.* MMNN/RF-LR alternates wide random-feature maps and low-rank bottlenecks. Symmetry is measured at the output, at bottleneck partials, and through mirror structure in first-layer atoms.

and apply the same relative defect to active internal partial functions. For first-layer ReLU atoms $\sigma(a_j^\top x + b_j)$, we also compare each atom to its nearest mirror $(-a_j, b_j)$ and measure outgoing coefficient mismatch.

## 2. Low-Rank Architecture

The architectural bias is not an explicit equivariance constraint. The first random-feature layer is sampled asymmetrically, and mini-batches contain a single point. Nevertheless, the trained low-rank channel weights can organize mirror-related atoms into an approximately symmetric representation.

## 3. Evidence

**One-dimensional ICLR reruns.** We reran the low-loss sumcos configurations from the ICLR landscape study with width 1024 and adaptive SGD. Across six reliable low-loss runs, final test MSE ranges from $1.94 \cdot 10^{-5}$ to $3.26 \cdot 10^{-3}$. Output even defects range from $4.1 \cdot 10^{-6}$ to $8.0 \cdot 10^{-4}$, while active last-layer partial even defects range from $6.24 \cdot 10^{-5}$ to $5.22 \cdot 10^{-4}$. This is the cleanest evidence that the learned symmetry is internal, not only an output coincidence.

**Counterexamples are useful.** The exhaustive rerun currently separates completed checkpoints into 12 partial-symmetric, 28 output-only/asymmetric, 92 intermediate, and 9 underfit runs. For example, one rank-5 depth-3 $f_2$ run

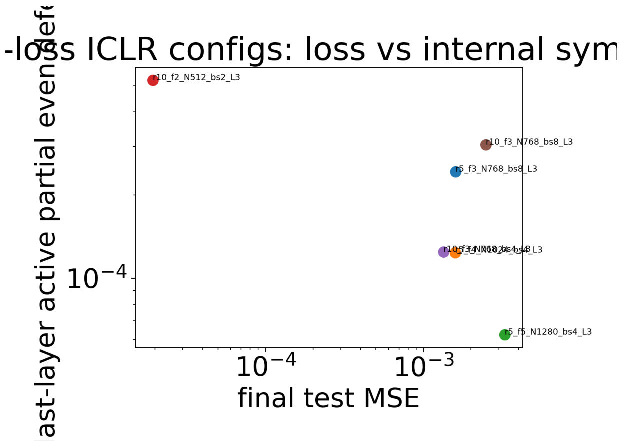


*Figure 2.* Low-loss ICLR sumcos reruns: fitted MMNNs have small output defects and small active last-layer partial defects.

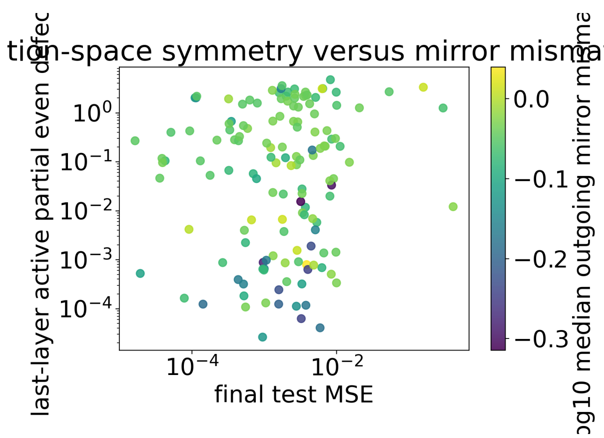


*Figure 3.* Completed reruns: output loss, partial symmetry, and median mirror mismatch reveal distinct regimes.

reaches test MSE $9.53 \cdot 10^{-4}$ with partial defect $2.62 \cdot 10^{-5}$. In contrast, a rank-10 depth-1 $f_1$ run reaches test MSE $1.17 \cdot 10^{-4}$ but partial defect 2.17. Thus low output loss does not force internal symmetry; the positive low-rank regime is a real representation phenomenon.

**Batch-one multidimensional stress test.** We then stress-test the strongest claim in two and three dimensions: symmetry can emerge from a fully asymmetric stochastic stream. All runs below use batch size one, non-paired uniform training samples, Adam, width 512, and low-rank bottlenecks with rank 8 or 16. The 2D Gaussian run with rank 8 and depth 3 reaches test MSE $1.20 \cdot 10^{-5}$, output defect $6.91 \cdot 10^{-5}$ under $x \mapsto -x$, and active last-partial defect $6.62 \cdot 10^{-4}$. The 3D quadratic run with rank 8 and depth 3 reaches test MSE $1.92 \cdot 10^{-4}$, output defect $9.76 \cdot 10^{-4}$, and active last-partial defect $1.81 \cdot 10^{-3}$. Thus the phenomenon survives beyond the original one-dimensional setting while using rank-to-width ratio $8/512 = 1.56\%$.

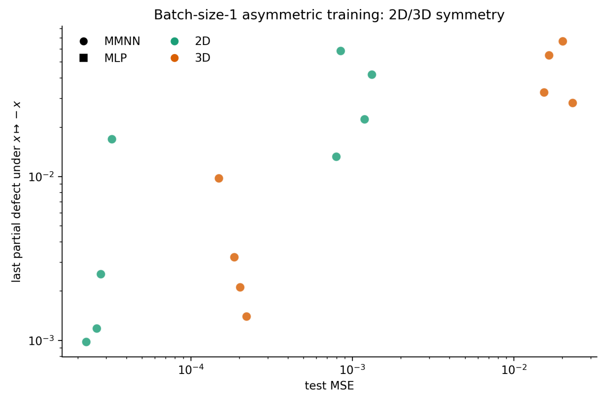


*Figure 4.* 2D/3D batch-size-one runs: low test error coincides with small active partial symmetry defects in the best low-rank regimes.

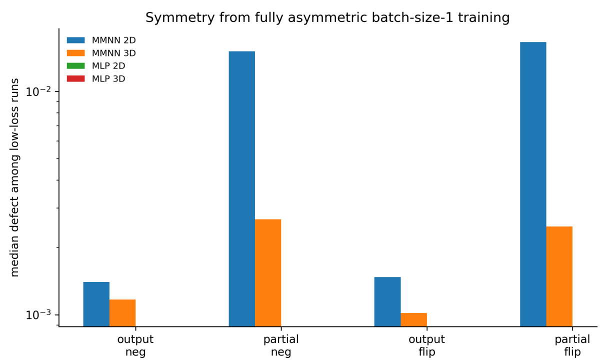


*Figure 5.* Median output and partial symmetry defects among low-loss multidimensional runs.

## 4. Why This Matters

The key message for weight-space symmetries is that useful symmetries need not be imposed as exact architectural equivariances or explicit path-norm constraints (Neyshabur et al., 2015). In low-rank random-feature networks, the optimizer can discover approximate, data-dependent symmetry in the trained low-rank weights. This suggests a practical analysis program: compare models not only by their functions, but by whether the symmetry appears in their internal partials and mirror-organized parameters.

## 5. Limitations

The current mirror metric is nearest-neighbor based and only probes the first layer. It supports the weight-space interpretation but is weaker than the partial-function evidence. Some harder 3D soft-axis targets still generalize poorly despite tiny training error, so the multidimensional claim should be stated for the low-loss Gaussian and quadratic regimes rather than as a universal statement across all symmetric targets.

## References


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